Fundamentals of Red Hat Enterprise Linux – Personal Notes

1. Vim Editor Basics

Saving and Quitting

:w → Save (write)

:wq → Save and quit

:q → quit without saving

ZZ → Save and quit (shortcut)

Copy and Paste

yy → Copy (yank) a line

p → Paste below the cursor

5p → Paste the copied line 5 times

Deleting Text

dd → Delete a line

4dd → Delete 4 lines

d$ → Delete from cursor to end of line

dw → Delete a word

Editing

cw → Change word

u → Undo last action

2. Visual Mode in Vim

v → Character visual mode

Shift + v → Line visual mode

Ctrl + v → Block visual mode

3. Useful Linux Commands

whoami → Displays the current logged-in user

command1 ; command2 → Executes both commands sequentially

command1 && command2 → Executes command2 only if command1 succeeds

echo $? → Shows return code of previous command

Return Codes:

0 → Command executed successfully

1 → Command failed

4. File and Text Commands

cat → Display contents of a file

head → Display first 10 lines of a file

tail → Display last 10 lines of a file

history → Show previously executed commands

5. Piping

command1 | command2

Output of command1 becomes input for command2

Example:

ls | grep file

6. Directory Commands

cd → Change directory

mkdir directory_name → Create a directory

mkdir -pv hello/hi/go → Create nested directories

ls -ld directory_name → Show directory details

7. System Monitoring

w → Shows users currently logged into the system

8. Running Applications in Background

firefox & → Runs program in background

9. Command Line Shortcuts

Esc + . → Recalls previous argument

10. User Management

usermod -g group_name username → Change user's primary group

11. Pattern Searching

grep pattern filename

grep stands for Global Regular Expression Print

Used to search for specific patterns in files

12. Adding a User

useradd username → Create a new user

passwd username → Set password for user

13. SSH Management

SSH (Secure Shell) is a secure protocol used to connect to another computer remotely.

Generate SSH key:

ssh-keygen

Copy SSH key to remote server:

ssh-copy-id user@remotehost

14. Linux Directory Structure

/ → Root directory

/home → User home directories

/root → Root user home

/bin → Essential commands

/sbin → System admin commands

/etc → Configuration files

/var → Logs and variable data

/tmp → Temporary files

/usr → Applications and utilities

/dev → Device files

15. File Permissions

Permissions control access to files.

r → read

w → write

x → execute

Example:

ls -l

-rwxr-xr--

User Group Others

Change permissions:

chmod 755 file.sh

Make file executable:

chmod +x script.sh

16. Changing File Ownership

Change owner:

chown user file.txt

Change owner and group:

chown user:group file.txt

17. Package Management (RHEL)

Install package:

sudo dnf install package_name

Remove package:

sudo dnf remove package_name

Update packages:

sudo dnf update

Search package:

dnf search package_name

18. Process Management

View running processes:

ps

ps aux

Real-time monitoring:

top

Kill process:

kill PID

Force kill:

kill -9 PID

19. Networking Basics

Check network interfaces:

ip a

Check connectivity:

ping google.com

Check open ports:

ss -tuln

20. Learning Resources

man command → Shows manual page

Example : man ls

vimtutor → Interactive vim tutorial